

Supplementary Materials: *Boosting Utility in Differential Privacy without Losing Privacy: the Maximum Knowledge Assumption*

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I. PROOFS

Proposition 1. *Let $f : \mathcal{D} \rightarrow \mathbb{R}^k$ be a query function, and let κ_f be an ϵ -DP mechanism for f . Let $U \subseteq \text{Range}(\kappa_f)$ be a subset representing the feasible output space. The mechanism κ'_f , obtained by reallocating the probability mass $\Pr(\kappa_f \notin U)$ entirely within U via a data-independent transformation, satisfies ϵ -differential privacy over the feasible space.*

Proof. To prove that κ'_f preserves ϵ -DP, we must show that for any measurable subset $V \subseteq \text{Range}(\kappa'_f)$ and for any pair of neighboring data sets $D, D' \in \mathcal{F}$, the following inequality holds:

$$\Pr(\kappa'_f(D) \in V) \leq e^\epsilon \Pr(\kappa'_f(D') \in V).$$

We split V into two disjoint parts based on the feasible space U : $V = (V \cap U) \cup (V \cap U^c)$.

1) Case 1: $V \cap U^c$ (Out-of-bounds region)

Because κ'_f reallocates all outside probability mass back into U , the probability of an output landing in the impossible complement space U^c is identically zero across all data sets:

$$\begin{aligned} \Pr(\kappa'_f(D) \in V \cap U^c) \\ &= 0 \\ &\leq e^\epsilon \cdot 0 \\ &= e^\epsilon \Pr(\kappa'_f(D') \in V \cap U^c) \end{aligned}$$

Thus, the privacy bound holds trivially.

2) Case 2: $V \cap U$ (Feasible region)

Let $\mu_U(V)$ be a data-independent, finite measure bounded by 1 that dictates how the escaped probability mass is distributed. Specifically, for any target range V , $\mu_U(V)$ defines the exact proportion of out-of-bounds mass reallocated to V

The output probability under this modified mechanism equals the sum of the retained core mass and the newly redistributed mass:

$$\begin{aligned} \Pr(\kappa'_f(D) \in V \cap U) \\ &= \Pr(\kappa_f(D) \in V \cap U) + \mu_U(V) \cdot \Pr(\kappa_f(D) \notin U) \end{aligned}$$

Since the underlying base mechanism κ_f is inherently ϵ -DP, both components independently respect our privacy parameters within the feasible space:

- For core mass: $\Pr(\kappa_f(D) \in V \cap U) \leq e^\epsilon \Pr(\kappa_f(D') \in V \cap U)$.
- For escaped mass: $\Pr(\kappa_f(D) \notin U) \leq e^\epsilon \Pr(\kappa_f(D') \notin U)$.

Multiplying the escaped mass inequality by our data-independent factor $\mu_U(V)$ yields:

$$\mu_U(V) \Pr(\kappa_f(D) \notin U) \leq e^\epsilon \mu_U(V) \Pr(\kappa_f(D') \notin U)$$

Combining these two components, we have:

$$\begin{aligned} \Pr(\kappa'_f(D) \in V \cap U) \\ &= \Pr(\kappa_f(D) \in V \cap U) + \mu_U(V) \Pr(\kappa_f(D) \notin U) \\ &\leq e^\epsilon \Pr(\kappa_f(D') \in V \cap U) + e^\epsilon \mu_U(V) \Pr(\kappa_f(D') \notin U) \\ &= e^\epsilon \Pr(\kappa'_f(D') \in V \cap U) \end{aligned}$$

Summing the results of Case 1 and Case 2 completes our proof:

$$\Pr(\kappa'_f(D) \in V) \leq e^\epsilon \Pr(\kappa'_f(D') \in V)$$

□

Proposition 2. *After removing the minimum and maximum values of X among the records that satisfy P within any valid neighbor $D' \in \mathcal{M}_D$, the values of X for all remaining records are guaranteed to fall within the empirical interval $[r'_1, r'_{n_P}]$.*

Proof. Because D' differs from the true data set D by at most one record, there can exist at most a single data entry resting outside the true empirical range $[r'_1, r'_{n_P}]$. Forcibly dropping

the minimum and maximum values ensures that all remaining entries fall within the boundaries of the original range. $\square$

Proposition 3. *The L_1 -sensitivity of the truncated mean under the maximum knowledge assumption evaluates to:*

$$\Delta \text{Mean}_P = \frac{r'_{n_P} - r'_1}{n_P - 3}.$$

Proof. Let $D', D'' \in \mathcal{M}_D$ be neighboring data sets. Denote by $|DS|_P$ the number of records in a data set DS that satisfy P . We know that $|D'|_P$ and $|D''|_P$ are between $n_P - 1$ and $n_P + 1$. We consider all possible combinations of values for $|D'|$ and $|D''|$, and compute the maximum change in the truncated mean in each case. The L_1 -sensitivity is the maximum among these. We have several cases:

- $|D'|_P = |D''|_P = n_P$. Both data sets differ by one record. By Proposition 2, the maximum change occurs when the differing record takes values r'_1 and r'_{n_P} , respectively. Since two extreme values are dropped, the mean is computed over $n_P - 2$ records:

$$\Delta = \frac{r'_{n_P} - r'_1}{n_P - 2}.$$

- $|D'|_P = |D''|_P = n_P - 1$. One fewer record satisfies P , so the truncated mean is computed over $n_P - 3$ values:

$$\Delta = \frac{r'_{n_P} - r'_1}{n_P - 3}.$$

- $|D'|_P = |D''|_P = n_P + 1$. One additional record satisfies P , so the truncated mean is computed over $n_P - 1$ values:

$$\Delta = \frac{r'_{n_P} - r'_1}{n_P - 1}.$$

- $|D'|_P = n_P, |D''|_P = n_P - 1$. This case corresponds to a record in D' satisfying P but not in D'' . The change in the truncated mean depends on whether the removed record is above or below the mean:
 - If the record in D' is below the mean, its removal in D'' increases the mean. However, the effect is smaller than when it is changed by r'_{n_P} .
 - If it is above the mean, its removal decreases the mean. However, the effect is smaller than when it is changed by r'_1 .

In any case, the effect is smaller than when changing a record and the maximum change is less than in the case $|D'|_P = |D''|_P = n_P$.

- $|D'|_P = n_P + 1, |D''|_P = n_P$. There is one extra record, and the maximum change is smaller than in the case $|D'|_P = |D''|_P = n_P + 1$.
- The remaining cases ($|D'|_P = n_P - 1, |D''|_P = n_P$; and $|D'|_P = n_P, |D''|_P = n_P + 1$), are symmetric with those already analyzed.

The maximum change occurs in case $|D'|_P = |D''|_P = n_P - 1$, where the truncated mean is computed over the fewest records, yielding the L_1 -sensitivity:

$$\Delta \text{Mean}_P = \frac{r'_{n_P} - r'_1}{n_P - 3}.$$

$\square$